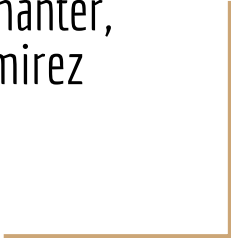


FSA BOEING

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Company - Boeing

- An American multinational corporation that designs, manufactures, and sells airplanes, rotorcrafts, rockets, satellites, and missiles worldwide. Their headquarters are in Arlington, Virginia.
- As a leading global aerospace company, Boeing develops, manufactures and services commercial airplanes, defense products and space systems for customers in more than 150 countries.
- Current Valuation \$179.79 Billion



Competitor 1: Lockheed Martin

- Like Boeing, Lockheed Martin is an American global aerospace, arms, and defense company . They were formed by the merger of Lockheed Corporation with Martin Marietta in March 1995.
- Lockheed Martin's main concentration are in aerospace products including aircrafts, space launchers, satellites, and defense systems. They are also a leading contractor for the U.S Department of Energy and NASA.
- \$461.91 Billion Valuation



Competitor 2: Raytheon



- Like Boeing and Lockheed, Raytheon Technologies is a American multinational aerospace company. Their headquarters are located in Arlington, Virginia.
- Raytheon is one of the main players in American Military Weaponry for land, sea, and air systems. They produce land launched missiles, aircraft radar systems, weapon sights, and targeting systems.
- Current valuation of \$101.40 Billion

Profitability

	Dec. 31, 2023	Dec. 31, 2022	
Sales	\$ 77,794		
NI	(2,242)		
Assets	137,012	137,100	
ATA	137,056		
Liabilites	154,240	152,948	
equity	(17,228)	(15,848)	
Average Equity	(16,538)		
NOPBT	(773)		
tax loss	(237)		
Tax shield	372		
NOPAT	(1,146)		
NOA			
Average NOA			

Profitability

DuPont ROE Disaggregation:

- ROE Boeing has negative NI and negative equity
- ROA reflects that Boeing is not profitable
- Negative FL reflects that Boeing does not have enough operating income to pay off debt

	Boeing	Raytheon	Lockheed
	Dec. 31, 2023	Dec. 31, 2023	Dec. 31, 2023
ROE	13.56%	4.71%	85.96%
ROA	-1.64%	1.99%	13.14%
FL	-8.29	2.37	6.54
PM	-0.03	0.05	0.1
AT	0.57	0.43	1.28
RNOA	NA	3.56%	34.37%
NOPM	NA	0.04	0.11
NOAT	NA	0.80	3.22

Profitability

DuPont Disaggregation of ROA

- PM: Boeing has a negative profit margin, because of its net loss. They lose 3 cents on each dollar sold
- AT: Boeing generates 57 cents on each dollar invested into assets
- RNOA
- Boeing has a negative NOPAT so results are inconclusive

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RNOA	NA	3.56%	34.37%
NOPM	NA	0.04	0.11
NOAT	NA	0.80	3.22

Asset Recognition

- Boeing's PPE Turnover Ratio was better compared to Raytheon but doesn't exceed Lockheed Martin's.
- This means that Boeing and Lockheed both utilize their fixed assets to generate revenue a little more compared to Raytheon which has a lower ratio.
- Boeing's Percent Used Up shows how much they are using their asset more than their competitors.
- Lockheed Martin's percent used up shows they are using their assets at a moderate rate while raytheon's show they are using their assets at a lower rate.

PPE Turnover Ratio = sales / average (net value of PPE)				
	Boeing	Raytheon	Lockheed Martin	
Sales	77794	68920	67571	
AVG PPE	10605.5	15459	8173	
PPE Turnover Ratio	7.34	4.46	8.27	
Percent Used Up				
	Boeing	Raytheon	Lockheed Martin	
Accumulated Deprec	-22245	-15644	-12812	
Avg Depreciable asset	15949.5	26026.5	18051	
Percent Used Up Ratio	139.47%	60.11%	70.98%	

Asset Recognition

Inventory Method

- Boeing - FIFO, Straight Line
- Raytheon - FIFO, Straight Line
- Lockheed Martin - LIFO, Straight Line
- We can see that Boeing has a lower inventory turnover compared to their competitors. While Raytheon and Lockheed Martin's inventory turnover indicates how efficient they might acquire and sell their inventory.

Inventory Turnover = COGS / average inventories				
	Boeing	Raytheon	Lockheed Martin	
COGS	-59864	43425	-59092	
Avg Inventories	78,946	11,197	3,110	
Inventory Turnover	0.76	3.88	19.00	

Ratios

Adjusted Equity to before share buyback*					
Ratios	Formula	Boeing	Lockheed Martin	Raytheon	
Current Ratio	Current Assets/Current Liabilities	1.14	1.21	1.04	
Quick Ratio	(Cash+short term investment + AR) / current liabilities	0.19	0.21	0.37	
Total Liability to equity	Total Liabilities / Equity	4.77	6.67	1.64	
Total Debt to Equity	(Long + Short Term Debt) / Equity	1.62	2.87	0.69	
Times interest Earned Ratio	EBIT / Interest Expense	0.31	9.29	2.37	
EBITDA Coverage Ratio	(EBIT+Depreciation & Amortization) / Interest Expense	-0.44	10.85	5.16	
Cash from Operation to Total Debt Ratio	Cash from Operations / (Short + Long Term Debt)	0.11	0.4	0.19	
Free Operating Cash Flow to Total Debt Ratio	Cash from Operations - Capex / Short Term + Long Term Debt	0.08	0.32	0.13	

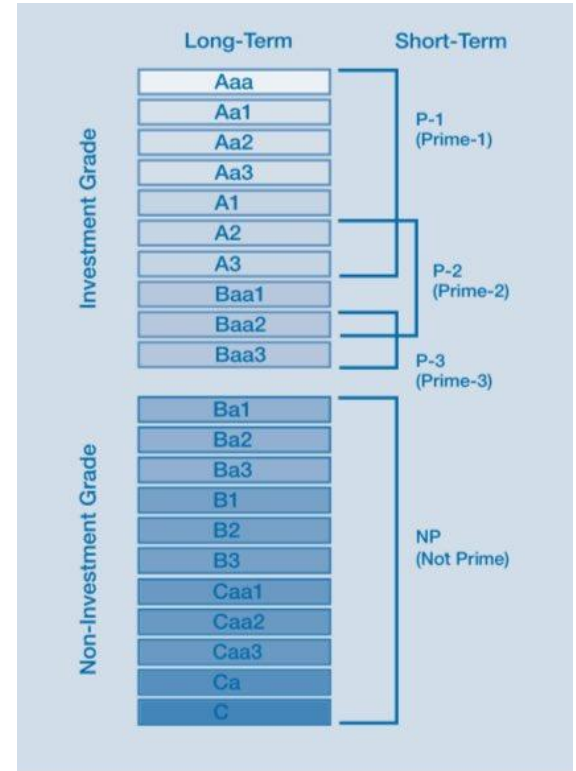
Brief discussion about boeing liquidity risk and solvency risk compared to competitors

- Due to Boeing's negative income which stems from the impact of COVID-19 and their 737 MAX crisis, they exhibit concerning solvency risks indicating liabilities and debt far exceed equity compared to Lockheed Martin, and Raytheon (-\$17,228,000 equity)
- Boeing Also faces significant liquidity risk with low quick and current ratios. This suggests a lack of sufficient liquid assets to cover short term liabilities. Comparably, Lockheed Martin exhibits relatively healthy ratios and Raytheon exhibits reasonable liquidity ratios
- In summary, Boeing faces severe solvency and liquidity risks according to these ratios compared to Lockheed Martin who is strong and healthy, and Raytheon who is moderately healthy.

Credit Ratings

*Based on Moodys Rating Scale

	Rating	Most Recent Affirm Date
<u>Boeing</u>	Baa3	4/29/2024
<u>Lockheed Martin</u>	A2	8/14/2023
<u>Raytheon</u>	Baa1	10/23/2023



2024 Forecast

Boeing Co	Year End 12/31					
(In Millions)	2023	% of Revenue	2024	% of Revenue	Variance	
Total Revenues	\$ 77,794		\$ 79,350		1,556	2.00%
Total Costs and expenses	\$ (70,070)	90.07%	\$ (71,471)	90.07%	1,401	2.00%
Gross Profit	\$ 7,724	9.93%	\$ 7,878	9.93%	154	2.00%
General and administrative expense	\$ (5,168)	6.64%	\$ (5,271)	6.64%	103	1.99%
Research and development expense, net	\$ (3,377)	4.34%	\$ (3,445)	4.34%	68	2.01%
Loss from operations	\$ (773)		\$ (788)		15	1.94%
Other income, net	\$ 1,227		\$ 1,252		25	2.00%
Income (loss) before income taxes	\$ (2,005)		\$ (1,996)		9	0.45%
Income tax (expense)/benefit	\$ (237)	0.35%	\$ (259)	0.33%	22	9.28%
Net Loss	\$ (2,242)	2.88%	\$ (2,255)	2.84%	13	0.58%

Equity analysis and Valuation

Boeing has negative Net income

The money multiple equations are going to give us inconclusive results.

Why is Boeing still an index giant?

Market Dominance

Long term government contracts

Economies of scale